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Hiding Behind Complexity: Supply Chain, Oversight, Race, and the Opioid Crisis

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Abstract

The opioid crisis has ravaged the United States, taking 69,000 lives in 2020, with prescription opioids accounting for 98% of opioid abuse. Although this epidemic is often considered a White public health crisis nationally, overdose deaths among people of color doubled from 2017 to 2019. As research has shown that the crisis was fueled by excessive supply from the pharmaceutical industry, several individual firms have received significant public criticism. However, we find evidence that the scope of the blame transcends individual actors to indict the very structure of complex supply chains, which may have exacerbated the crisis by dispensing significantly more opioids. In specific, we posit that supply chain complexity allowed mass quantities of opioids to escape detection by the Drug Enforcement Administration (DEA). Further, we find new evidence showing the greater impact of complexity on opioid dispensing in non-White communities, which underscores their exclusion from the public discourse and governmental response surrounding the crisis and suggests possible racial bias in the DEA's regulatory policies. Our analysis was made possible by the 2019 release of the DEA's Automation of Reports and Consolidated Orders System database, which logged every shipment in the US opioid supply chain from 2006 to 2014. Using a fixed effects model, we find that a one-unit increase across three dimensions of supply chain complexity is associated with a 16% increase in opioid dispensing. This effect is intensified in non-White communities, where a 10% increase in the non-White population is associated with a 3.39% (1.33%) increase in opioid dispensing by pharmacies that have supply chains with high (average) complexity. To verify that high-complexity pharmacies' excess dispensing supplied non-medical/recreational demand, we exploit the reformulation of OxyContin (designed to prevent recreational use) as an exogenous shock to the market. In a novel approach, we leverage the fact that different pharmacies received their first shipment of reformulated OxyContin at different times and use a difference-in-differences model to estimate the heterogeneous effect of the shock on dispensing. As the reformulated OxyContin stifled demand, high-complexity pharmacies experienced a 15.31% greater reduction in dispensing compared to lower-complexity pharmacies, suggesting that their excess dispensing was indeed satisfying non-medical/recreational demand.

Keywords

Opioid epidemic, supply chain, complexity, racial bias

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> CVS, Walgreens, Giant Eagle, and Walmart failed to stop mass quantities of opioid drugs from reaching the black market, fueling hundreds of overdose deaths amid one of the worst public health crises in the nation's history.
>
> —The Washington Post, October 3, 2021

1 Introduction

Scholars argue that the excessive supply of prescription opioid drugs is the primary cause of the opioid crisis in the United States, and blame has been cast on all the individual players in the opioid supply chain for helping to fuel it (Maclean et al., 2020). Manufacturers aggressively marketed prescription opioids and paid doctors to prescribe unnecessary dosages (Kornfield et al., 2022), distributors continued to supply the drugs despite clear signs of suspicious ordering

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from their customers (Bernstein et al., 2016), and pharmacies failed to develop adequate controls to prevent drugs from being diverted into illicit markets (Lopes and Kornfield, 2022). The ample evidence of pharmaceutical companies' contribution to the opioid crisis has resulted in thousands of lawsuits that have already generated billions of dollars in settlements (Achenbach et al., 2019).

In the face of the heated debate surrounding individual companies' roles in the opioid crisis, we argue that beyond the culpability of individual actors, the structural characteristics of the prescription opioid *supply chains* may have created an environment conducive to the excessive supply of opioids. Furthermore, we report evidence indicating that contrary to media portrayals where "it seems the majority of the victims [. . .] are white" (Shihipar, 2019), non-White communities were more acutely affected by supply chain mechanisms that facilitated the excessive proliferation of opioids.

But how could pharmaceutical companies push exorbitant quantities of opioids through a distribution system that is—ostensibly—heavily regulated by the Drug Enforcement Administration (DEA)? The DEA is tasked with monitoring the prescription opioid supply chain, which involves collecting data on all the shipments of controlled substances through the Automation of Reports and Consolidated Orders System (ARCOS) and requiring manufacturers and distributors to report suspicious orders of unusual size and frequency.We argue that the complexity of pharmaceutical supply chains, defined as the number and variety of the supply chain members and their interactions (Choi and Krause, 2006), prevents the DEA from effectively monitoring prescription opioid supply—a challenge that has been acknowledged by the US government itself: "[The] DEA is ill-equipped to effectively monitor ordering patterns for all pharmaceutical opioids" (Office of the Inspector General, 2019). Figure 1 highlights the difference between simple and complex supply chains.

Parallel to the DEA's inadequate monitoring efforts, drug enforcement escalated in non-White communities, highlighting a longstanding gap in equitable treatment (Rosino and Hughey, 2018): "[P]harmaceutical companies . . . whittled away regulatory guardrails . . . [while] the 'war on drugs' mindset handles addicted people like criminals" (Macy, 2022). Governmental agencies, including the DEA, have a well-documented history of discrimination against communities of color (Pincus, 2019), including disproportionate arrests for drug possession and trafficking (Donnelly et al., 2021) and inadequate provision of social services (Potapchuk et al., 2005). The opioid crisis is yet another example: The government's framing of the opioid epidemic as both a "drug crisis" and a "public health crisis" has created a racial double standard (James and Jordan, 2018), and the exclusion of non-White communities from the social and political discourse surrounding the opioid crisis has implications for the public health response in these communities. We contend that the focus on White rural communities may have resulted in an unequal allocation of resources for monitoring opioid distribution that exacerbated the influence of supply chain complexity on over-dispensing in non-White communities. Figure 2 frames the structure and impact of this system on vulnerable communities.

As supply chains become more complex, they become difficult for companies to manage due to the increasingly intensive information processing required to monitor activities (Skilton and Robinson, 2009; Galbraith, 1973). We recognize a similar phenomenon in DEA regulatory efforts by studying horizontal, vertical, and spatial complexities that capture the breadth, depth, and geographical spread of the opioid supply chain (Sharma et al., 2020; Giannoccaro et al., 2018; Bode and Wagner, 2015).

Research on supply chain structures is hindered by the scarcity of reliable granular data on connections and interactions in supply networks (Akın Ateş et al., 2022; Sharma et al., 2020). Challenges include a lack of longitudinal supply data (Lu and Shang, 2017), limited coverage of the firms in the supply network (Gualandris et al., 2021), and confinement to using only publicly disclosed data sources such as Compustat and Bloomberg supply chain analysis (SPLC) database (e.g., annual reports and press releases) to map complex relationships in supply chains (Sharma et al., 2020). The public release of ARCOS in 2019 overcame many of these challenges, providing rich data for accurately measuring complexity in the opioid supply chain; its massive dataset enabled us to analyze millions of prescription opioid transactions. For each transaction, we observe the product, supplier, customer, transaction date, and quantity ordered for 90,925 pharmacies, 1,290 distributors, and 870 manufacturers.

Our results suggest that a one-unit increase in the horizontal and spatial complexities is associated with a 2.12% and 14.45% increase in opioid dispensing, respectively, although we observe a slight drop associated with vertical complexity. Furthermore, a 10% increase in the non-White proportion of a county results in a 3.39% (1.33%) increase in opioid dispensing by pharmacies that have supply chains with high (average) complexity.

A logical question is whether the excessive supply is due to legitimate medical needs or non-medical/recreational demand. To answer this, we exploit the exogenous shock of OxyContin's 2010 reformulation to prevent its recreational use. Previous literature indicates that this reformulation reduced recreational demand for OxyContin without affecting medical demand (Evans et al., 2019). As the recreational demand was stifled, pharmacies with high supply chain complexity saw a 15.31% greater drop in dispensing after reformulation relative to pharmacies with low complexity.

Contributions. Researchers have previously explored the influence of individual actors at different levels of the supply chain on mass opioid distribution by studying payments from manufacturers to physicians (Hadland et al., 2019), the effect of state policies on the progression of the crisis toward illicit opioid drugs (Kc et al., 2022), and the role of pharmacy ownership in excessive opioid dispensing (Janssen and

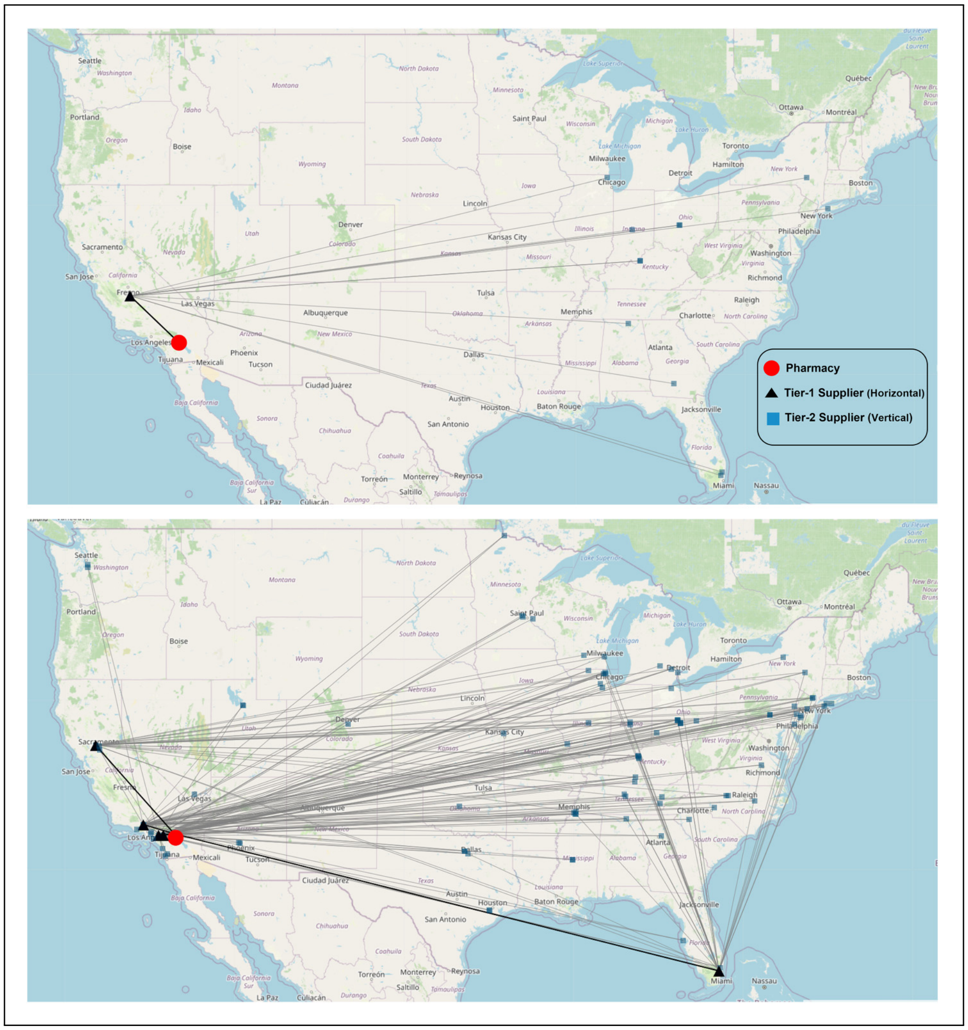

Figure 1. Supply chain structures of a Walmart (top panel) and a Walgreens pharmacy (bottom panel) in Riverside Co., CA.

Zhang, 2023). Our paper contributes to the growing empirical literature on factors affecting prescription opioid supply by bringing a new perspective that shows how responsibility goes beyond individual actors, whereby the broader *supply chain* network may have allowed large shipments of opioids to reach consumer markets undetected by the DEA. Specifically, we show that a significant factor in unregulated dispensing derives from the *interactions* between manufacturers, distributors, and pharmacies in a supply network.

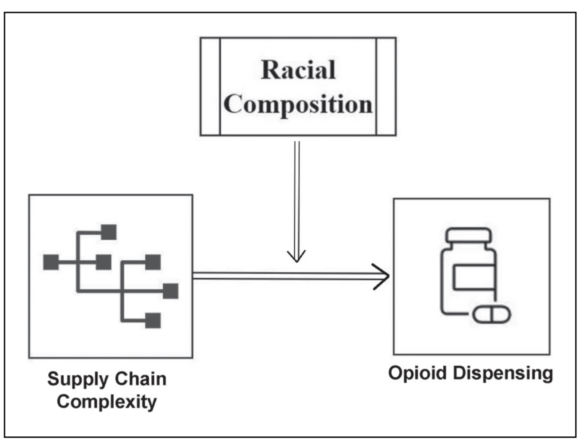

Figure 2. Conceptual illustration of the impact of disparate Drug Enforcement Administration (DEA) oversight on the relationship between supply chain complexity and opioid dispensing.

Prior studies have focused on supply chain complexity's *negative impact* on *business performance*, such as financial performance (Lu and Shang, 2017), innovation (Chu et al., 2019; Vickery et al., 2016), and the probability of disruptions (Bode and Wagner, 2015). By contrast, we consider the perspective of *regulators*, as opposed to the firm, and show that complexity may have facilitated the excessive supply of opioid drugs by complicating the task of DEA oversight.

Finally, we contribute to the literature on bias and discrimination in the fields of Operations Management (OM) and opioid epidemic. Most OM scholars have focused on biases from individuals' perspectives, such as employers' discrimination in hiring (Kuppuswamy and Younkin, 2020; Acquisti and Fong, 2020), drivers' biases in accepting ride requests (Mejia and Parker, 2021), and discrimination by hosts against guests in online marketplaces (Cui et al., 2020). By contrast, we take a societal perspective, focusing on systematic racial bias and its nationwide effect on the government's lopsided response to the opioid epidemic. We spotlight the impact of the crisis on communities of color, providing evidence that supply chain complexity has a greater influence on excess dispensing in these communities, potentially due to weaker efforts at oversight. In doing so, we expose the fallacy that the opioid crisis is an exclusively White health crisis—a fallacy driving a public health response that largely ignores non-White communities.

2 Background and Related Literature

Our research lies at the intersection of three streams of literature: prescription opioid supply, supply chain complexity, and racial disparities in response to the opioid crisis.

2.1 Prescription Opioid Supply

2.1.1 The Excessive Supply of Prescription Opioids. Prescription opioids are widely regarded as the root cause of the opioid crisis (Alpert et al., 2022). In 2020, 9.3 million of the 9.5 million people abusing opioids reported having misused prescription pain relievers (Substance Abuse and Mental Health Services Administration, 2022). Previous studies argued that the excessive supply of prescription opioids was the primary contributor to the crisis (Maclean et al., 2020). At the height of the opioid crisis in 2012, there were 81.3 prescriptions per 100 persons in the US (Paulozzi et al., 2014).

Previous literature has explored factors such as promotional activities by drug manufacturers (Mejia et al., 2019), physician's prescribing behavior (Bobroske et al., 2022; Schnell and Currie, 2018), and insurers' coverage of opioids (Zhou et al., 2016). We argue that the important question is not only *which actors* contributed to the excessive distribution of opioids but also *how* they could do so in a tightly regulated market. Our contribution lies in showing that supply networks behave differently from their individual actors by demonstrating the impact of *interactions* between pharmaceutical firms on facilitating excessive opioid distribution.

To the best of our knowledge, only Skilton and Bernardes (2022) have studied the opioid crisis from a supply chain perspective. They suggest that supply pool pressure, measured at the state level, results in excessive dispensing by pharmacies. By contrast, we study *complexity* at the *pharmacy level* and consider connections between pharmacies, distributors, and manufacturers.

2.1.2 The Role and Responsibility of Pharmacies in the Opioid Crisis. Pharmacies, as the gatekeepers of the prescription opioid supply, play a crucial role in curbing opioid abuse and drug diversion. All pharmacists are tasked with ensuring that medications are dispensed for legitimate medical purposes; they must assess prescription validity and monitor patient behavior for signs of misuse or abuse (Bach and Hartung, 2019; DEA, 2022). In such cases, pharmacists have the authority to dispense only partial quantities or even refuse to fill a prescription (DEA, 2022). However, pharmacies are implicated in ~80% of prescription drug diversion cases that involve prescription theft or forgery, insurance fraud, and similar malfeasance (Inciardi et al., 2007). Interviews involving pharmacists and substance abusers reveal significantly varied degrees of gatekeeping compliance in handling suspicious prescriptions by different pharmacists (Hartung et al., 2018), and numerous recent legal rulings and settlements against pharmacies provide direct evidence of their failure to monitor and control the distribution of prescription opioids (e.g., Kornfield 2021; Achenbach et al. 2019). A notable example involves a recent lawsuit against Consumer Value Stores (CVS), Walgreens, and Walmart—three of the largest pharmacy chains in the US—wherein they agreed to pay $650.5 million in settlements for their role in fueling the opioid epidemic in Ohio (Lopes and Kornfield, 2022). The lawsuit revealed that Walgreens and Walmart denied pharmacists access to crucial information for diversion control, while CVS withheld tools that could have alerted pharmacists to "red flags." Furthermore, pharmacists' bonuses were tied to prescription volume, instilling fear that addressing red flags or refusing to fill prescriptions could negatively impact their pay. Walgreens explicitly told its pharmacists that "[the] best evidence of a well-run pharmacy is the increase in prescriptions and pharmacy sales" (Trumbull County, OH, et al. v. Purdue Pharma L.P., et al., 2022).

2.2 Supply Chain Complexity and Oversight

2.2.1 DEA Monitoring of the Prescription Opioid Supply Chain. The DEA monitors the supply chain of controlled substances using a multi-pronged strategy that strongly emphasizes manufacturers' and distributors' proactive detection and reporting of suspicious orders. Under the Controlled Substances Act, entities such as manufacturers, distributors, and pharmacies must register with the DEA, thus ensuring accountability and facilitating communication between the agency and the supply chain actors (DEA, 2010). The DEA also mandates that shipments of controlled substances be reported to the ARCOS database, enabling tracking and analysis of these substances through the supply chain. Crucially, the DEA requires distributors and manufacturers to closely monitor their customers' ordering patterns, holding them responsible for identifying and reporting any suspicious orders that deviate from the norm in terms of size or frequency. Figure 3 provides an overview of how the DEA seeks to regulate the prescription opioid supply chain.

2.2.2 Challenges Posed by Supply Chain Complexity. Most definitions of supply chain complexity center on Herbert Simon's description of a complex system as "a system that includes a large number of varied elements that interact in a non-simple way" (Simon, 1991). A key theory by Simon (1991) posits that systems behave differently from individual parts due to

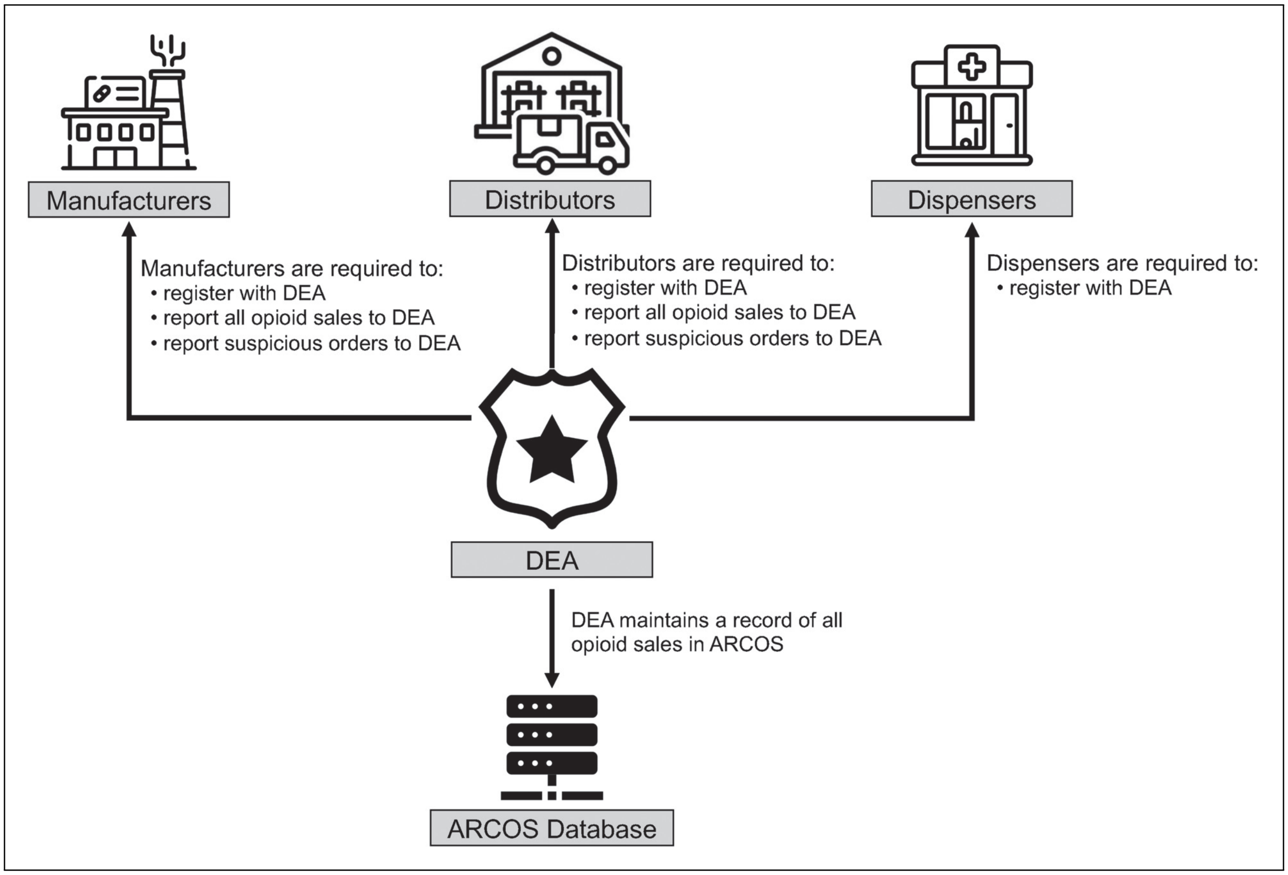

Figure 3. Visual overview of how the Drug Enforcement Administration (DEA) oversees the prescription opioid supply chain.

complex interactions, making it difficult to infer the whole system's behavior from the behavior of its parts. Supply chain networks are also viewed as complex systems with interrelated components (Choi et al., 2001). This systems perspective allows us to uncover new insights into the drivers of the opioid crisis.

Research has shown that supply chain complexity decreases operational and financial performance (Fan et al., 2022; Wang et al., 2021; Lu and Shang, 2017; Bozarth et al., 2009), increases the likelihood of disruptions (Phadnis and Joglekar, 2021; Bode and Wagner, 2015), and hinders innovation (Chu et al., 2019; Vickery et al., 2016; Bellamy et al., 2014). One primary reason is that increasing the number and diversity of supply network members increases the difficulty of *control and oversight* (Dong et al., 2020; Sharma et al., 2020; Giannoccaro et al., 2018; Lu and Shang, 2017; Skilton and Robinson, 2009). We contribute to the knowledge in this domain by arguing that, by similar logic, supply chain complexity hindered DEA oversight of prescription opioid distribution and thus allowed large quantities of opioids to pass undetected through the supply network. Horizontal complexity refers to the number of direct (Tier-1) suppliers (Bode and Wagner, 2015; Bozarth et al., 2009; Choi and Hong, 2002), vertical complexity represents the number of Tier-2 suppliers per Tier-1 supplier (Sharma et al., 2020; Lu and Shang, 2017), and spatial complexity measures the geographical diversity of direct suppliers (Bode and Wagner, 2015; Lu and Shang, 2017) (see Figure 4).

Supply chain complexity poses significant challenges to the DEA's monitoring efforts because it can exploit gaps in the regulatory system. One key limitation in the DEA's monitoring approach is that suppliers and manufacturers lack access to their customers' full ordering history—that is, they can only track pharmacy orders that they fulfill themselves. Because horizontal complexity involves ordering from multiple suppliers that typically don't share data, a suspiciously large shipment can be broken into multiple smaller shipments, with each individual supplier receiving a smaller, less suspicious order. Similarly, by ordering from multiple suppliers, a pharmacy can order at a higher frequency without being detected. This "workaround" constitutes a direct mechanism by which supply chain complexity may influence the supply of opioids and subvert regulatory efforts.

The DEA is divided into 23 field divisions based on geography—another vulnerability in its monitoring scheme that can be exploited by supply chain complexity. These divisions operate independently and lack a unified system for consolidating and sharing information, which hampers

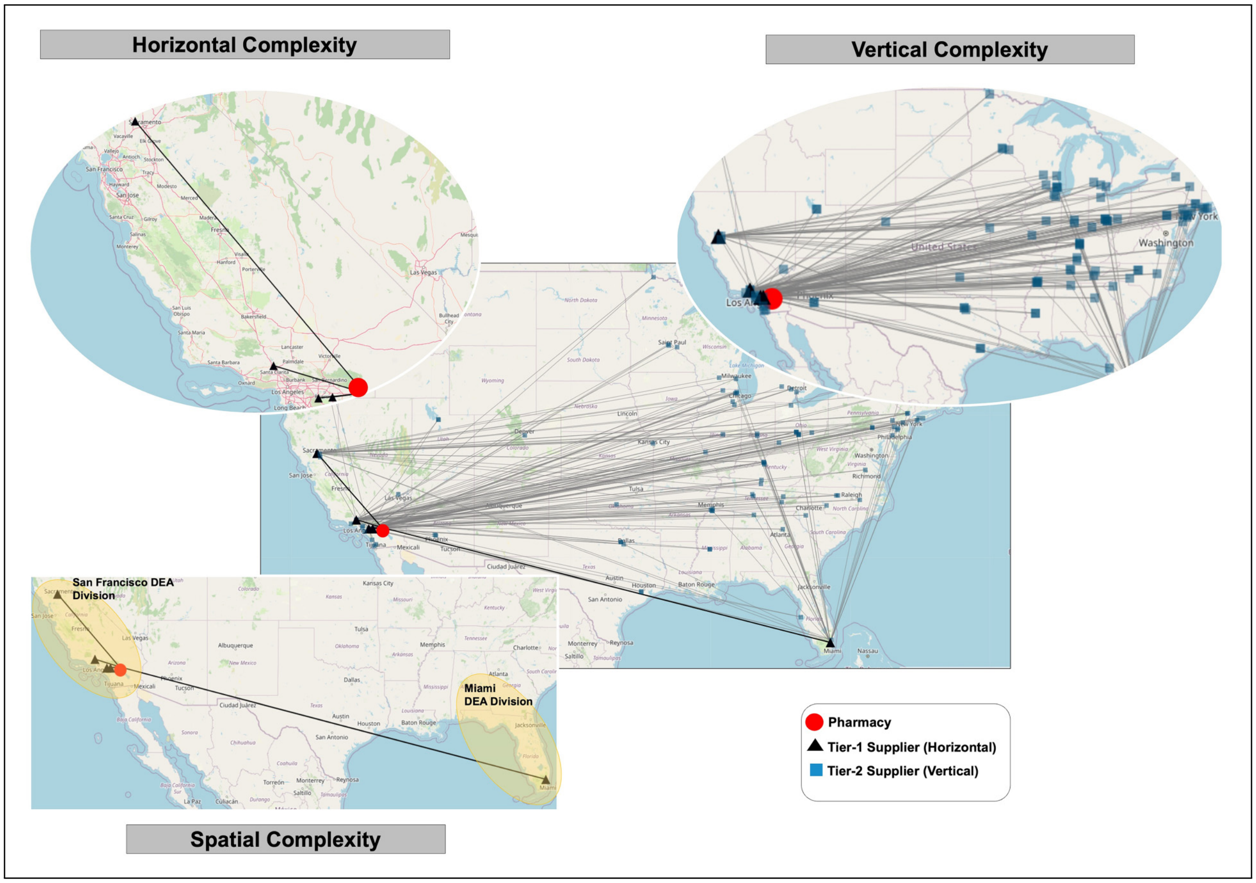

Figure 4. Example of supply chain complexity dimensions.

effective cross-referencing and coordination (U.S. Government Accountability Office, 2020). Spatial complexity, which involves ordering from different geographical regions, can exploit this lack of communication between DEA divisions. For example, even if a supplier reports a suspicious order in one division, other divisions that the pharmacy orders from are unlikely to be informed. A recent Energy and Commerce Committee report on the opioid crisis in West Virginia identified the lack of communication between different DEA divisions as a potential factor contributing to inconsistency in the DEA's responses to suspicious orders: "Due to the decentralized nature of reporting, there exists a possibility that the DEA's 23 field divisions may review or investigate suspicious orders in an inconsistent manner" (The Energy and Commerce Committee, 2018). In general, fragmentation of orders makes it challenging for individual suppliers, manufacturers, or DEA divisions to detect irregularities because the true scale of a pharmacy's ordering is obscured by the complex web of transactions.

Much of the extant literature proposes solutions for how *firms* at different levels of a supply chain can improve their performance outcomes by managing the complexity of their supply networks. Chu et al. (2019) found that suppliers can improve innovation by increasing their proximity to their customers, while Bode and Wagner (2015) suggested that companies can reduce the frequency of disruptions by simplifying their supply chains. We add to this stream by examining how supply chain structures impact *regulators* as well.

2.3 Racial Discrimination in Response to the Opioid Crisis

The growing interest in examining bias and discrimination in business has generated studies of racial bias in healthcare delivery (Ganju et al., 2020), ridesharing platforms (Mejia and Parker, 2021), inequality in hiring (Kuppuswamy and Younkin, 2020; Fernandez-Mateo and Fernandez, 2016), and fairness in decision-making algorithms (De-Arteaga et al., 2022). This burgeoning domain presents significant opportunities to expand our knowledge about the influence of bias and discrimination on broader aspects of operations and supply chain management (Corbett et al., 2022; Jónasson et al., 2022). We contribute to this literature by exploring racial disparities in regulation that have created previously unnoticed opportunities for excessive opioid dispensing by combining supply chain complexity with lackluster oversight in communities of color.

Racial discrimination in response to the opioid crisis has been well documented by previous studies, which reveal a concerning pattern of under-allocating resources and support to non-White communities (Andraka-Christou, 2021; Hart and Hart, 2019; James and Jordan, 2018). A notable example of this disparity is the limited access to treatment services, where inequity in minority groups' access to opioid addiction treatment options, such as medication-assisted treatment, is well documented in the literature (Hollander et al., 2021). Chang et al. (2022) found that hospitals in communities with high proportions of Black or Hispanic populations were less likely to offer the most common hospital-based programs to address opioid use disorder. Furthermore, buprenorphine and naloxone—safer alternatives to methadone for the treatment of opioid use disorder—are less accessible than methadone in minority communities (Santoro and Santoro, 2018).

We argue that this pattern of disparity may extend to the DEA's allocation of surveillance resources in the opioid crisis. Like other government entities, the DEA's efforts may have been influenced by racial biases surrounding the response to the opioid crisis. For example, the Office of the Inspector General recently reported that the DEA is seeking to resolve a staffing shortage by increasing Diversion Investigator and Special Agent staffing in field divisions hardest hit by the opioid epidemic, which popular opinion holds to be predominantly White communities (Office of the Inspector General, 2019). This evidence leads us to hypothesize that the positive relationship between supply chain complexity and opioid dispensing is exacerbated in counties with higher proportions of non-White residents.

3 Data and Empirical Strategy

3.1 Data

Our study combines data about pharmacies with county-level demographic and socioeconomic factors. We extract opioid dispensing and supply chain structures of pharmacies from ARCOS. Previously, ARCOS was publicly available only as state-level summary reports; however, a year-long litigation spearheaded by The Washington Post and Charleston Gazette-Mail in 2019 resulted in the public release of the full ARCOS database for 2006–2014, providing a unique opportunity for more comprehensive studies of the prescriptions opioid supply chain (Achenbach, 2019).

Reports in the popular press restrict their ARCOS analyses to transactions of oxycodone and hydrocodone pills, thereby eliminating 30% of the available data and potentially biasing the results. We extend this work by including all opioid drugs except methadone and buprenorphine—which are used in medication-assisted treatment (MAT) of opioid dependence—and excluding transactions with hospitals, clinics, and practitioners. Following the economic and public health literature (e.g., Powell et al. 2020), we measure supply in morphine milligram equivalents (MMEs) instead of pill counts to account for the potency of distributed drugs.

We construct a panel dataset at the pharmacy-year level containing 584,610 observations of 90,925 pharmacies in the US from 2006 to 2014, combining it with population, age, racial, gender compositions, median household income, and poverty rate data from the two Census Bureau datasets, unemployment rates from the Local Area Unemployment Statistics, and the number of substance abuse treatment facilities per 100,000 population from the U.S. Census Bureau's County Business Patterns.[1]

3.2 Variables

3.2.1 Dependent Variable. We use *opioid dispensing* measured in millions of MMEs as our dependent variable based on compelling academic evidence of the positive association between medical access to prescription opioids and adverse opioid outcomes (Bohnert et al., 2011). Powell et al. (2020) showed that a 10% increase in the legal supply results in a 7.1% increase in opioid-related overdose deaths and that $\sim 75\%$ of this increase in deaths (between 2000 and 2011) resulted from diversions of prescription opioids to illicit markets, which was facilitated by the dramatic increase in opioid supply from the pharmaceutical industry.

3.2.2 Independent Variables. We consider three dimensions of supply chain complexity—horizontal, vertical, and spatial—as our main independent variables. We measure horizontal complexity as the number of Tier-1 suppliers directly connected to the focal firm (Sharma et al., 2020; Bode and Wagner, 2015). Leveraging the richness of the ARCOS dataset, we are able to extend this measure by incorporating the strength of the pharmacy–supplier relationship based on the quantity of prescription opioids (in MMEs) shipped from each Tier-1 supplier to the focal pharmacy. To quantify supply dispersion, we use the Gini coefficient, which is widely employed to measure heterogeneity (Park and Berenguer, 2020). $HCOMP_{it}$ calculates the horizontal complexity of pharmacy i in year t:

$$HCOMP_{it} = N_{it} + G_{it}(1 - N_{it}), \tag{1}$$

where N_{it} is the total number of suppliers of pharmacy i in year t, and G_{it} is the Gini coefficient.

We measure vertical complexity as the number of Tier-2 suppliers per Tier-1 supplier (Sharma et al., 2020; Lu and Shang, 2017) weighted by the strength of the connection of each Tier-1 supplier with pharmacy i in year t, $\frac{d_{jit}}{d_{it,Total}}$, where d_{jit} represents the MMEs distributed from Tier-1 supplier j to pharmacy i in year t, and $d_{it,Total}$ denotes the total MMEs distributed to pharmacy i in year t, $d_{it,Total} = \sum_{j=1}^{N_{it}} d_{jit}$. If we let $Tier1\ HCOMP_{jt}$ be the adjusted number of suppliers of each Tier-1 supplier from equation (1), vertical complexity

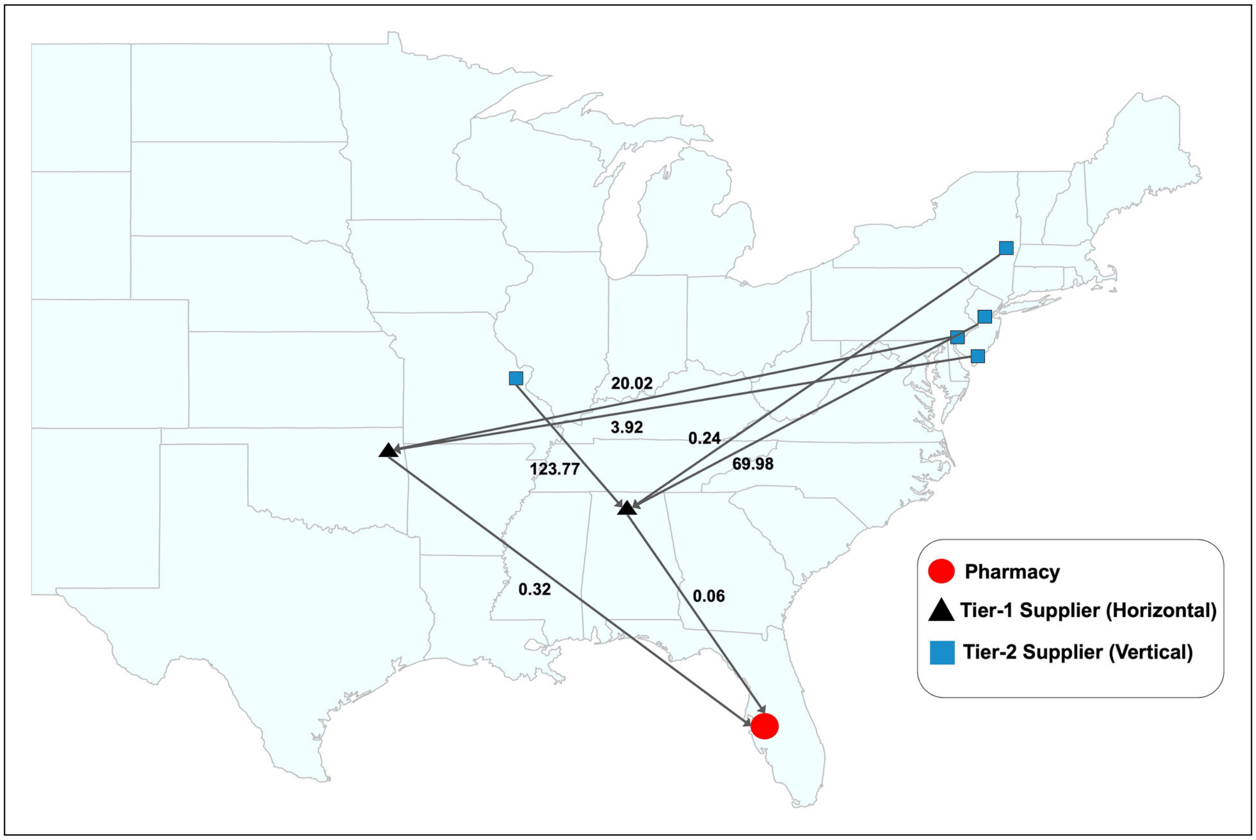

Figure 5. Example of how complexity measures are calculated. The numbers next to the edges in the lines represent the quantity of opioids (in million MMEs) shipped between vertices. The supply dispersion among the two Tier-1 suppliers is calculated using the Gini coefficient, which is equal to 0.68. Thus, $HCOMP = 2 + 0.68(1 - 2) = 1.32$. For vertical complexity, we first calculate the horizontal complexities of each Tier-1 supplier, which are equal to 1.33 and 1.72. Then, we have $VCOMP = 0.84 \times 1.33 + 0.16 \times 1.72 = 1.39$. $SCOMP = 2$ because Tier-1 suppliers are located in two different DEA divisions. DEA: Drug Enforcement Administration; MMEs: morphine milligram equivalents; HCOMP: horizontal complexity; VCOMP: vertical complexity; SCOMP: spatial complexity.

is

$$VCOMP_{it} = \sum_{j=1}^{N_{it}} \frac{d_{jit}}{d_{it,Total}} \times Tier1\ HCOMP_{jt}. \qquad (2)$$

We measure spatial complexity, $SCOMP_{it}$, as the number of different DEA divisions (23 total) in which Tier-1 suppliers of pharmacy i are located. Similar measures have been used for the geographical diversity of supply chains (Lu and Shang, 2017). Descriptive statistics and correlation between variables are provided in E-Companion, Section EC.1. Figure 5 provides an example of how the complexity measures are calculated. Figure 6 provides model-free evidence of the association between complexity and opioid dispensing, and Figure 7 compares the patterns of opioid dispensing and non-White proportion in California counties with the same level of supply chain complexity.

3.3 *Empirical Analyses and Results*

Our main analysis leverages an OLS model with pharmacy and year fixed effects (FEs) to control for time-invariant confounding factors and national trends that might affect opioid dispensing. Following the literature, we also include state-year FEs to account for state-level policies affecting opioid distribution (Hollingsworth et al., 2017). The model is specified as

$$\begin{aligned} log(opioid\ dispensing_{it}) = {} & \beta_1 HCOMP_{it} + \beta_2 VCOMP_{it} \\ & + \beta_3 SCOMP_{it} + \beta_4 Controls_{it} \\ & + \delta_i + \eta_t + \mu_{st} + \varepsilon_{it}, \end{aligned} \qquad (3)$$

where $opioid\ dispensing_{it}$ is the amount of prescription opioids dispensed by pharmacy i in year t; δ_i, η_t, and μ_{st} are pharmacy, year, and state-year FEs, respectively; and ε_{it} is the error term. The log transformation is used to reduce skewness. We include an extensive set of county-level demographic and socioeconomic controls used in previous studies to account for time-varying confounding factors (Kc et al., 2022; Liu and Bharadwaj, 2020; Nguyen et al., 2019; Hollingsworth et al., 2017). We also include the number of pharmacies located in the same ZIP code as the focal pharmacy to control for the potential impact of competition on dispensing levels.

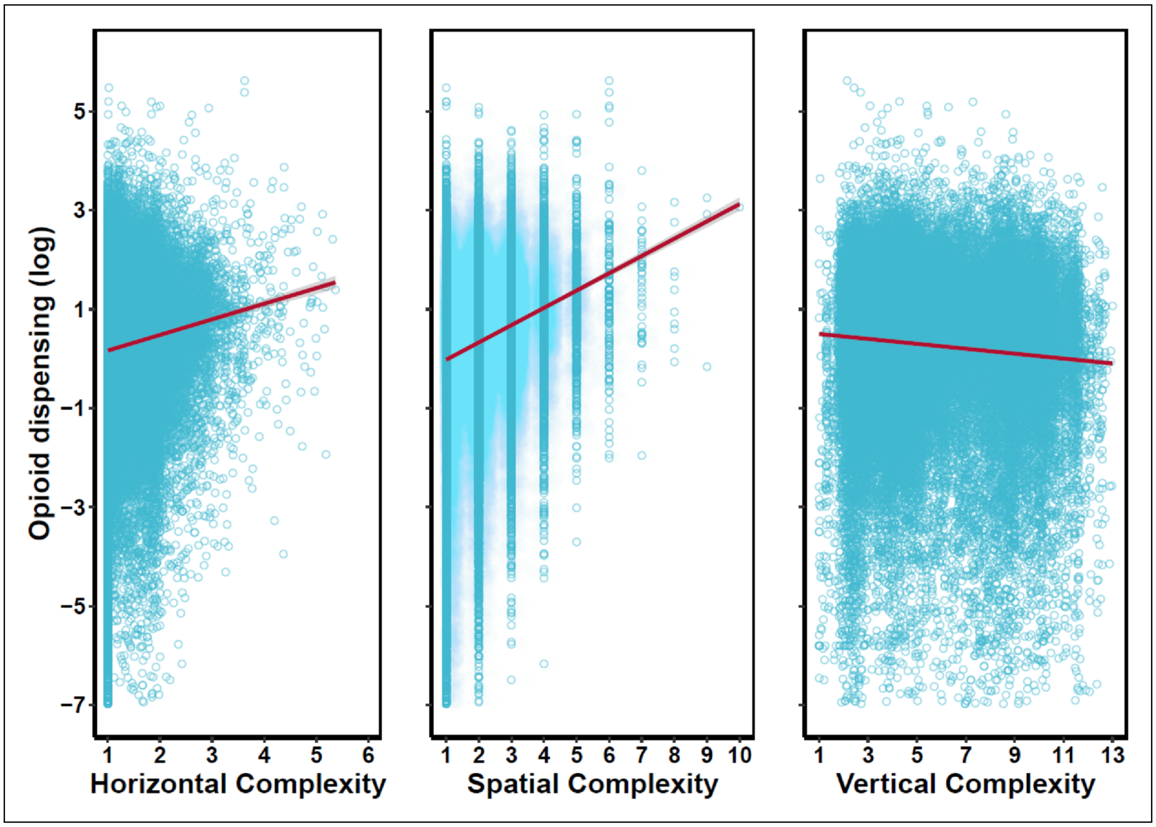

Figure 6. Scatter plots of opioid dispensing (million morphine milligram equivalents (MMEs)) versus supply chain complexity dimensions.

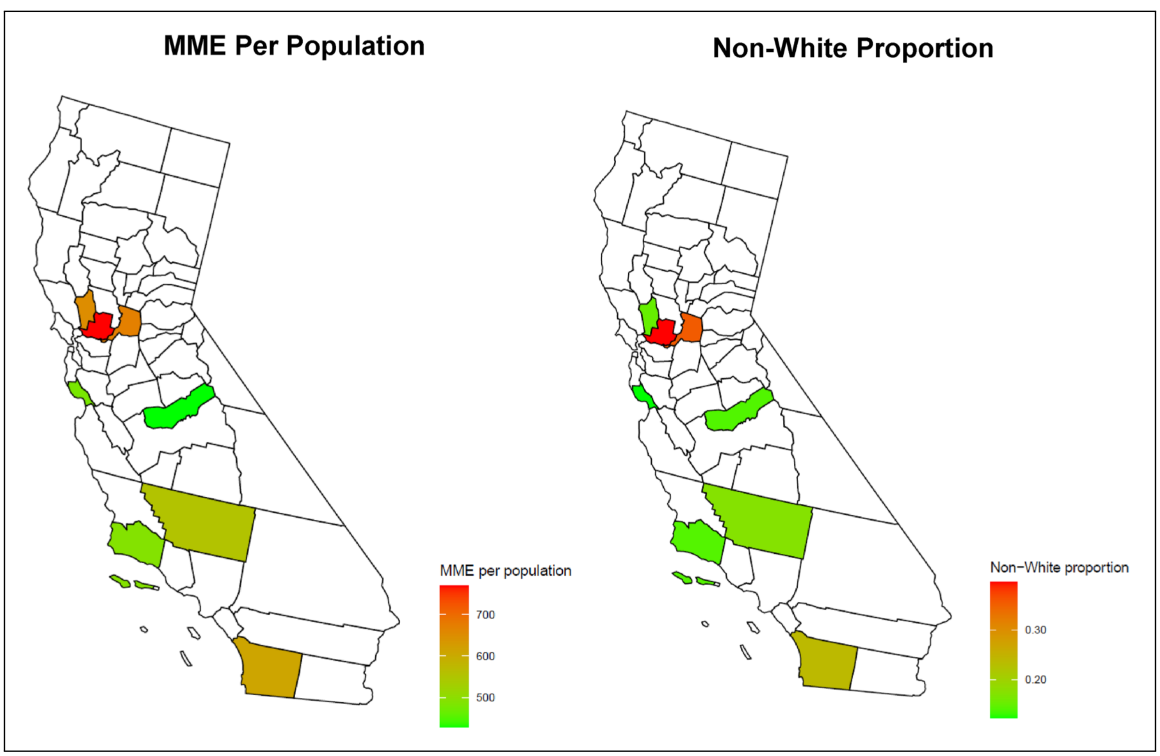

Figure 7. Patterns of opioid dispensing and non-White proportion in counties with the same level of supply chain complexity.

We found no variance inflation factors greater than five among our variables, thus mitigating multicollinearity concerns (Kutner et al., 2004). Table 1 provides the estimation results for equation (3). Column (1) shows the effect of supply chain complexity on opioid dispensing without controls. In columns (2) and (3), we add time-varying controls and state-year FEs, finding that the estimation is robust to these factors. In all models, standard errors are clustered at the pharmacy level to account for serial correlation. The estimated effect of supply chain complexity is significant for all three dimensions at the 1% level. A one-unit increase in the horizontal and spatial complexities is associated with 2.12%[2] and a 14.45% increases in opioid dispensing, respectively. A one-unit increase in vertical complexity is associated with a small decrease (0.6%) in opioid dispensing.

As previously discussed, greater horizontal and spatial complexities can increase dispensing by impeding DEA oversight, and the significant positive coefficients for these complexities support this argument. Interestingly, vertical complexity has a negative (albeit *much smaller*) impact on dispensing. This heterogeneity in the effects of complexity dimensions—especially for vertical complexity—aligns with past studies indicating that Tier-1 suppliers exert a more immediate and substantial influence on a company's performance than those deeper within the network (Lu and Shang, 2017; Wilhelm et al., 2016). Vertical complexity's less pronounced impact is due to its more distant influence and weaker direct connection to oversight, relative to horizontal and spatial complexities (for additional supporting arguments, see E-Companion, Section EC.2).

One potential concern regarding our estimation is that pharmacies might increase the complexity of their supply chains in response to heightened opioid demand. However, given the financial and operational drawbacks associated with supply chain complexity, this seems improbable (for a more in-depth discussion on this topic, see E-Companion, Section EC.3).

To examine the moderating impact of racial composition, we estimate the following model:

$$\begin{aligned} log(opioid\ dispensing_{it}) = {} & \beta_1 HCOMP_{it} + \beta_2 VCOMP_{it} \\ & + \beta_3 SCOMP_{it} + \beta_4 HCOMP_{it} \\ & \times NonWhite_{it} + \beta_5 VCOMP_{it} \\ & \times NonWhite_{it} + \beta_6 SCOMP_{it} \\ & \times NonWhite_{it} + \beta_7 Controls_{it} \\ & + \delta_i + \eta_t + \mu_{st} + \varepsilon_{it}, \end{aligned} \tag{4}$$

with $NonWhite_{it}$ representing the proportion of non-White residents of the county where pharmacy i is located in year t. Column (4) in Table 1 presents the estimation results. The positive and significant coefficients of interaction terms $HCOMP \times NonWhite$ ($\beta = 0.070,\ p < 0.01$) and $SCOMP \times NonWhite$ ($\beta = 0.081,\ p < 0.01$) indicate that as non-White proportion increases, both horizontal and spatial complexities have a stronger effect on pharmacies' opioid dispensing. The interaction term $VCOMP \times NonWhite$ is negative and statistically significant ($\beta = -0.025,\ p < 0.01$). These results indicate that for a pharmacy with high (average) supply chain complexity, a 10% increase in the non-White proportion of the population increases opioid dispensing by 3.39% (1.33%).

4 Robustness Checks

To resolve the concern that our primary analysis cannot distinguish between legitimate medical and non-medical/recreational demand, we exploit the 2010 OxyContin reformulation into an abuse-deterrent formula as a quasi-experiment to separate the two demand streams. Previous studies have shown that medical demand for OxyContin remained the same while non-medical use was reduced by up to 40% after reformulation (Cicero and Ellis, 2015). Figure 8 shows the overall impact on OxyContin dispensing, whereas other opioids were not affected (see E-Companion, Figure EC.2).

We test whether pharmacies with higher complexity experience a sharper drop in OxyContin dispensing after reformulation due to the loss of non-medical/recreational demand using two different identifications:

Differences Model: Following previous literature (e.g., Alpert et al. 2018), we assume all pharmacies experienced the reformulation shock simultaneously. We then separate a subset of pharmacies into high- and low-complexity groups and compare the relative effect of reformulation on their dispensing.

Difference-in-Differences (DiD): Using a novel approach, we identify the heterogeneous timing of the shock by considering when each pharmacy first received reformulated OxyContin and use a DiD specification.

4.1 Differences Model

We apply k-means clustering on three dimensions of complexity and use the elbow method to determine that the optimal grouping for pharmacies is four clusters (see E-Companion, Section EC.5 for details). Table 2 shows the average complexity dimensions in each group. We choose pharmacy groups 1 and 4 as high- and low-complexity groups, respectively.

To test our hypothesis, we estimate the following equation:

$$\begin{aligned} OxyContin\ dispensing_{it} = {} & \beta_1 High\ Complexity_i \times Post_t \\ & + \beta_2 Controls_{it} + \delta_i + \eta_t + \mu_{st} + \varepsilon_{it}, \end{aligned} \tag{5}$$

where i represents pharmacy, and t represents year. $OxyContin\ dispensing_{it}$ is measured in million MMEs, and $Post_t$ is an indicator variable that equals 1 for all years after 2010.

Figure 9 compares the average OxyContin dispensing of pharmacies with high versus low complexity before and after reformulation. Before reformulation, there was a significant gap between pharmacies in high and low groups. After

Table 1. Effect of supply chain complexity on opioid dispensing.

	Log(opioid dispensing)			
	(1)	(2)	(3)	(4)
Horizontal Complexity	0.009**	0.010**	0.021***	0.003
	(0.004)	(0.004)	(0.004)	(0.007)
Vertical Complexity	−0.004***	−0.006***	−0.006***	0.0004
	(0.001)	(0.001)	(0.001)	(0.002)
Spatial Complexity	0.132***	0.131***	0.135***	0.117***
	(0.002)	(0.002)	(0.002)	(0.004)
log(Population)		0.655***	0.641***	0.634***
		(0.071)	(0.086)	(0.086)
Unemployment Rate		0.010***	0.003	0.002
		(0.001)	(0.002)	(0.002)
Poverty Rate		0.002**	0.001	0.001
		(0.001)	(0.001)	(0.001)
Median Household Income		−0.005***	−0.003***	−0.003***
		(0.001)	(0.001)	(0.001)
Male Proportion		−2.539***	−0.523	−0.571
		(0.908)	(0.927)	(0.927)
Non-White Proportion		−1.885***	−1.824***	−1.959***
		(0.327)	(0.360)	(0.366)
Hispanic Proportion		−2.517***	−0.744*	−0.690*
		(0.341)	(0.403)	(0.402)
Age 15–29 Proportion		0.490	0.613	0.567
		(0.409)	(0.458)	(0.458)
Age 65+ Proportion		−0.752*	0.168	0.243
		(0.416)	(0.461)	(0.460)
Competition		−0.044***	−0.042***	−0.042***
		(0.001)	(0.001)	(0.001)
No. of Treatment Facilities per 100,000 Population		−0.003***	−0.001	−0.001
		(0.001)	(0.001)	(0.001)
Horizontal Complexity×Non-White Proportion				0.070***
				(0.024)
Vertical Complexity×Non-White Proportion				−0.025***
				(0.007)
Spatial Complexity×Non-White Proportion				0.081***
				(0.014)
Pharmacy FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
State-Year FEs	No	No	Yes	Yes
Observations	584,606	550,832	550,832	550,832
R^2	0.863	0.866	0.868	0.868
Adjusted R^2	0.838	0.841	0.842	0.842

Note: FEs: fixed effects. Standard errors in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.
Standard errors are clustered at the pharmacy level.

reformulation, we observe a sharp decline in OxyContin dispensed by high-complexity pharmacies, eventually rendering the two groups indistinguishable.

Table 3 presents the results of estimating equation (5). In our preferred specification, column (4), the negative coefficient of the term, *High Complexity* × *Post*, indicates that high-complexity pharmacies experienced a 0.049 million MME (15.31%) reduction in dispensing relative to low-complexity pharmacies. This suggests that the excess supply observed in higher-complexity pharmacies in our primary analysis (Table 1) was satisfying non-medical/recreational OxyContin demand. We validate that these differences are not due to pre-existing trends before reformulation in E-Companion, Section EC.6. Moreover, we check the robustness of the results with respect to the number of clusters and the method of clustering in E-Companion, Section EC.7.

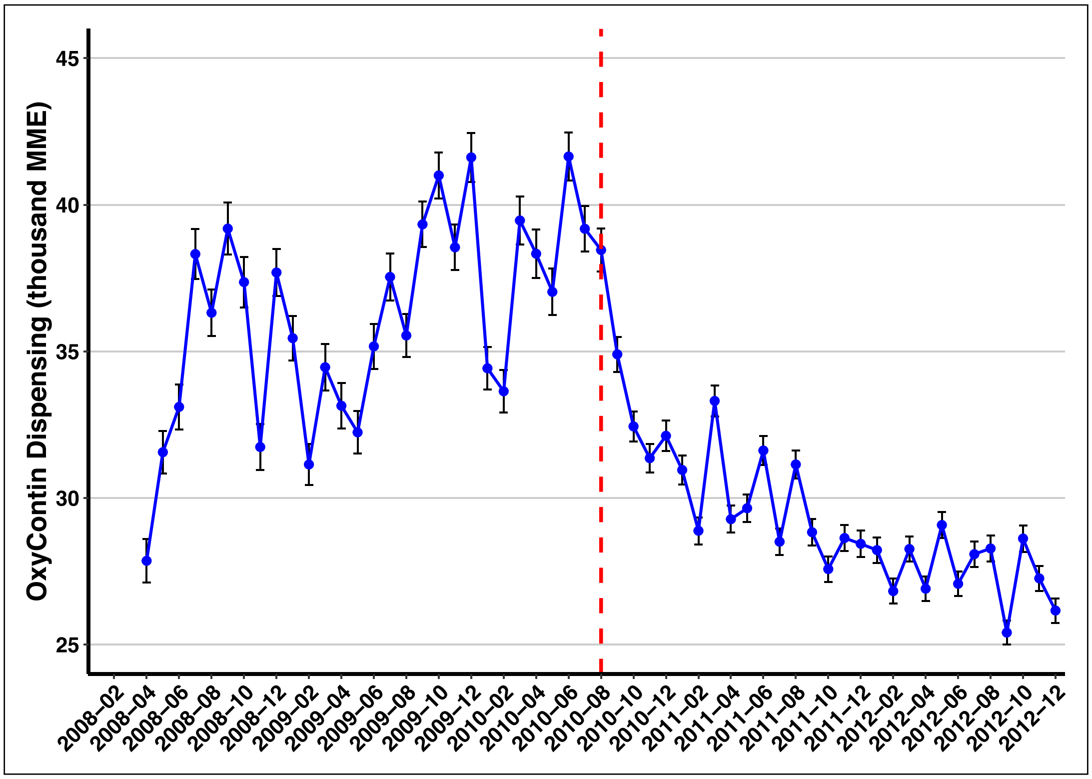

Figure 8. Monthly average dispensing of OxyContin by pharmacies from 2008 to 2012 before and after OxyContin reformulation in August 2010 (red dotted line). The error bars correspond to the 95% intervals.

Table 2. Mean of supply chain complexity dimensions.

		Supply chain complexity		
Group	*N*	Horizontal	Vertical	Spatial
1	7,864	2.085	5.050	3.361
2	22,388	1.602	3.533	2.222
3	15,516	1.142	9.059	1.621
4	21,831	1.137	3.759	1.296

4.1.1 Effect of Racial Composition. Next, we examine the moderating effectof racial composition on the greater reduction in OxyContin dispensing observed in high-complexity pharmacies after reformulation. Figure 10 demonstrates that the relative reduction in OxyContin dispensing in pharmacies with high complexity was greater in counties with higher non-White proportions. This suggests that more non-medical/recreational dispensing was occurring in non-White communities from pharmacies with higher complexity, supporting our primary analysis.

4.2 DiD Framework

One concern in previous studies arose from the assumption that reformulated OxyContin was distributed to all pharmacies on the same date, where the lack of variation in the timing of the shock eludes a DiD specification. To our knowledge, ours is the first study to overcome this barrier using the exact date (year-month) when pharmacies received their first shipment of reformulated OxyContin as the time of the treatment. Heterogeneous timing in receiving reformulated OxyContin (Figure 11) yields multiple treatment and control groups.

We construct *monthly* panel data of OxyContin dispensing at the pharmacy level and estimate the following DiD model:

$$\begin{aligned} \textit{OxyContin dispensing}_{it} = {} & \beta_1 Post_{it} + \beta_2 HCOMP_i^{(2008-2009)} \\ & \times Post_{it} + \beta_3 VCOMP_i^{(2008-2009)} \\ & \times Post_{it} + \beta_4 SCOMP_i^{(2008-2009)} \\ & \times Post_{it} + \delta_i + \eta_t + \mu_{st} + \varepsilon_{it}, \quad (6) \end{aligned}$$

where $HCOMP_i^{(2008-2009)}$, $VCOMP_i^{(2008-2009)}$, and $SCOMP_i^{(2008-2009)}$ measure pre-reformulation horizontal, vertical, and spatial complexities, respectively, of pharmacy i averaged over 2008–2009. $Post_{it}$ equals 1 for all months after pharmacy i received reformulated OxyContin and 0 otherwise.

Columns (1) and (2) of Table 4 use the full sample, and columns (3) and (4) limit the sample to 2009–2011 to more precisely estimate the effect around the time of reformulation. The moderation effects of horizontal and vertical complexities are consistently significant and negative across all models. However, the moderation effect of spatial complexity is significant only when we limit the sample to 2009–2011. The greater

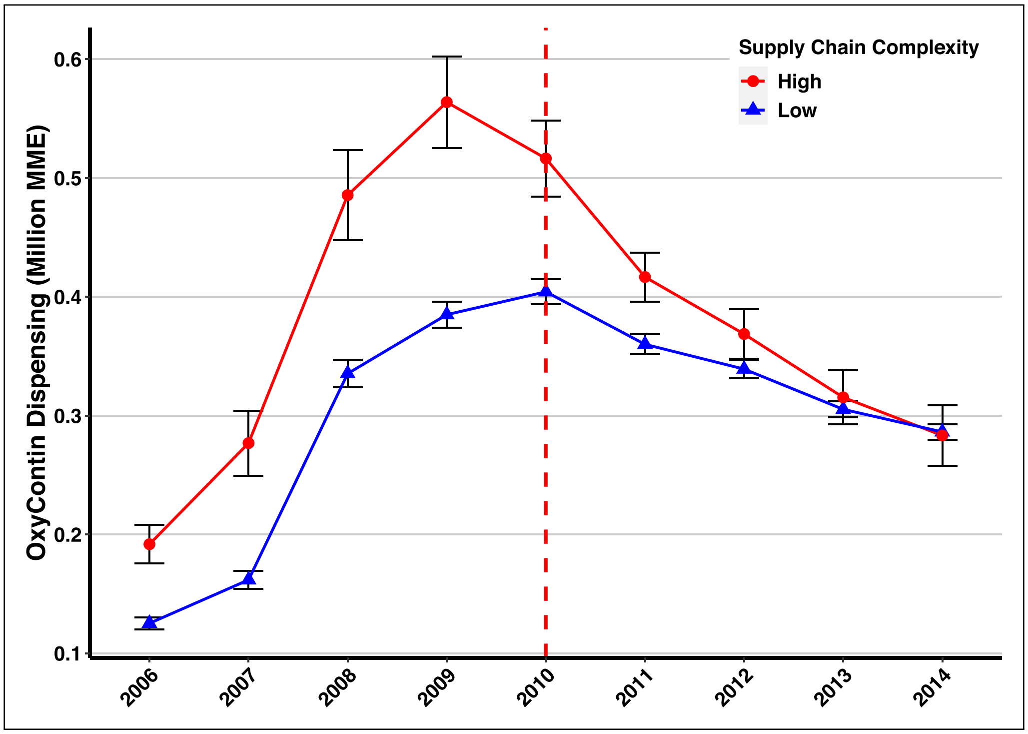

Figure 9. Average dispensing of OxyContin by pharmacies with high versus low supply chain complexity before and after OxyContin reformulation in August 2010 (red dotted line). The error bars correspond to the 95% intervals.

Table 3. OxyContin reformulation regression results.

	OxyContin dispensing			
	(1)	(2)	(3)	(4)
High Complexity	0.160***	0.155***		
	(0.016)	(0.016)		
Post	0.009			
	(0.006)			
High Complexity × Post	−0.091***	−0.088***	−0.047***	−0.049***
	(0.013)	(0.013)	(0.011)	(0.012)
Constant	0.068			
	(0.223)			
Controls	Yes	Yes	Yes	Yes
Year FEs	No	Yes	Yes	Yes
Pharmacy FEs	No	No	Yes	Yes
State-Year FEs	No	No	No	Yes
Avg. OxyContin Dispensing[a]	0.32	0.32	0.32	0.32
Avg. Effect (%)[b]	28.44	27.50	14.69	15.31
Observations	219,210	219,210	219,210	219,210
R^2	0.020	0.024	0.679	0.683
Adjusted R^2	0.020	0.024	0.630	0.634

Note: FEs: fixed effects; MMEs: morphine milligram equivalents. Standard errors in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.
Standard errors are clustered at the pharmacy level.

[a] Mean of the outcome variable, OxyContin dispensing, in million MMEs per pharmacy.

[b] The percentage reduction in the difference between OxyContin dispensing of high and low-complexity pharmacies. It is calculated by dividing the estimated coefficient of *High-complexity* × *Post* by the average OxyContin dispensed (in million MMEs) per pharmacy.

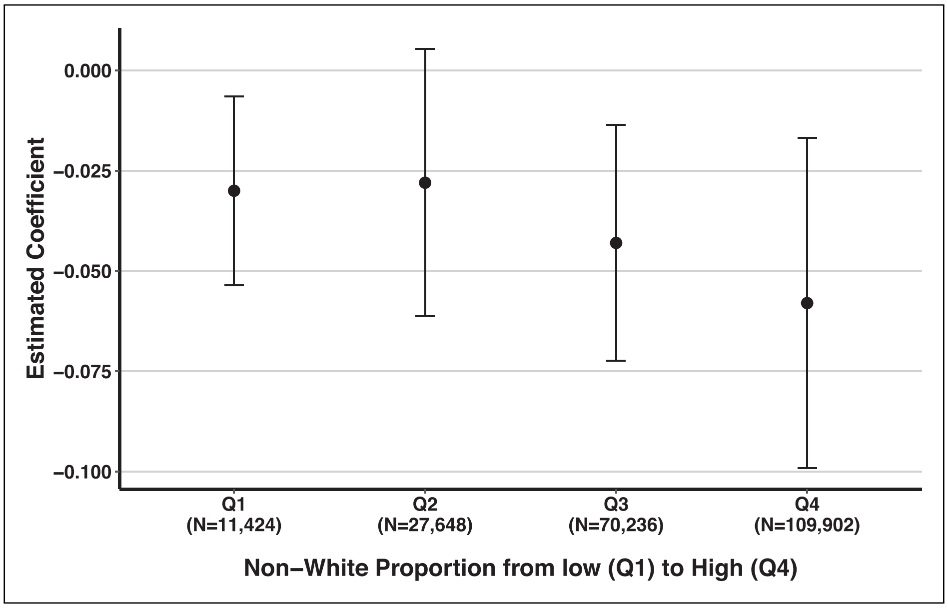

Figure 10. Each point depicts the estimated coefficient of the interaction term in column (4) of Table 3. Pharmacies are partitioned into four groups based on the 25th, 50th, and 75th quartiles of the non-White proportion. The number of observations in each group is displayed. The error bars correspond to the 95% intervals.

Table 4. Moderation effects of supply chain complexity measures.

	OxyContin dispensing			
	Full sample: 2006–2014		Subsample: 2009–2011	
	(1)	(2)	(3)	(4)
Post	0.020***	0.018***	0.015***	0.016***
	(0.002)	(0.002)	(0.002)	(0.003)
HCOMP×Post	−0.005***	−0.004***	−0.003**	−0.003*
	(0.001)	(0.001)	(0.001)	(0.001)
VCOMP×Post	−0.001***	−0.001***	−0.001***	−0.002***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
SCOMP×Post	−0.0003	−0.001	−0.002***	−0.003***
	(0.0005)	(0.001)	(0.001)	(0.001)
Pharmacy FEs	Yes	Yes	Yes	Yes
Year-Month FEs	Yes	Yes	Yes	Yes
State-Year FEs	No	Yes	No	Yes
Observations	6,088,611	6,087,012	2,126,513	2,124,914
R^2	0.623	0.626	0.763	0.764
Adjusted R^2	0.619	0.622	0.756	0.757

Note: HCOMP: horizontal complexity; VCOMP: vertical complexity; SCOMP: spatial complexity; FEs: fixed effects. Standard errors in parentheses.
$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.
Standard errors are clustered at the pharmacy level.

relative reduction in OxyContin dispensing observed in pharmacies with higher supply chain complexity further supports our hypothesis that the excessive supply from these pharmacies was likely driven by non-medical/recreational demand.

Reformulated OxyContin shipments arrived at different times to each pharmacy. In these circumstances, the heterogeneous impact of the treatment over time can introduce bias in the estimation of the treatment effect when using a standard two-way FE method (Goodman-Bacon, 2021; De Chaisemartin and d'Haultfoeuille, 2020). To confirm the robustness of our results, we also estimate the DiD specification using the methods suggested by Gardner (2022) and Callaway and Sant'Anna (2021) (see E-Companion, Section EC.8). Furthermore, to address the concerns regarding potential endogeneity in the timing of pharmacies' receipt of the reformulated OxyContin, we employ the synthetic control

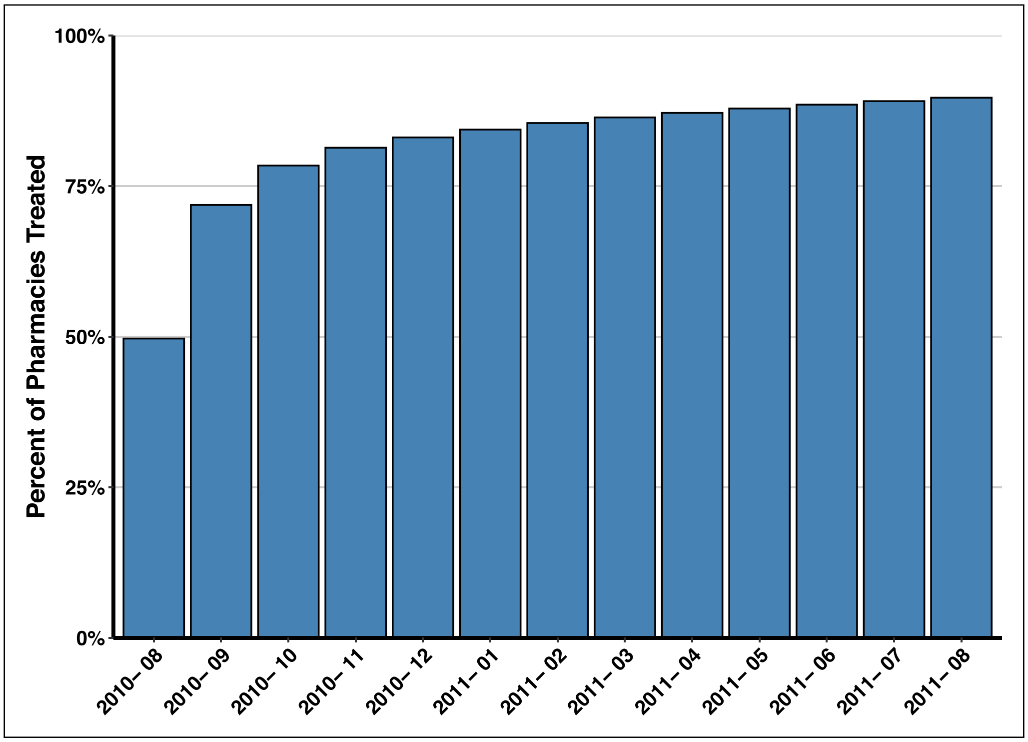

Figure 11. Percentage of pharmacies that received reformulated OxyContin in a one-year period after August 2010.

DiD method developed by Arkhangelsky et al. (2021) (see E-Companion, Section EC.9). The results from these alternative methods are consistent with our previous findings.

4.3 Autonomy in Supplier Selection

According to our data, retail pharmacies vary widely in their supplier choices—however, there is a potential concern that individual pharmacies within national chains (e.g., Walgreens and CVS) may have limited independence in choosing their suppliers under a centralized management system. However, a detailed examination of ARCOS data reveals notable differences in supply chain structures, even among individual stores of the same pharmacy chains. Figure 12 depicts considerable variation in complexity measures within major US pharmacy chains; these significant variations within the pharmacy chains can also be observed in specific states. For instance, Figure 13 shows the distribution of complexity measures for pharmacies in Florida, a state with a significant presence of chain pharmacies. These findings suggest that individual pharmacies, or their decision-makers, have some level of independence in choosing their suppliers.

5 Discussion

This study explains how, despite the DEA's efforts to regulate the opioid supply, pharmaceutical companies were able to saturate the US with massive quantities of opioids even as the opioid crisis spun out of control. Although the crisis is nationally regarded as a White public health crisis, overdose deaths tripled in Black communities from 2014 to 2020 (Substance Abuse and Mental Health Services Administration, 2020). Our work is timely, given growing concerns about the DEA's failure to control the diversion of prescription opioids and the utter lack of attention given to communities of color (U.S. Government Accountability Office, 2020; Office of the Inspector General, 2019).

Our research uncovers how supply chain complexity may have facilitated the influx of exorbitant quantities of opioids to the market undetected by the DEA. Our FE model showed that a one-unit increase in pharmacies' horizontal and spatial complexities is associated with 2.12% and 14.45% increases in opioid dispensing, respectively. Our novel DiD approach, which uses OxyContin reformulation as an exogenous shock in 2010, adds credence to our results. These findings have important implications for the DEA's monitoring efforts: Rather than focusing solely on individual transactions between distributors and pharmacies, the DEA should adopt a holistic supply network perspective to oversee and regulate the distribution of prescription opioids more effectively.

Furthermore, we show that supply chain complexity had a stronger association with the increase in opioid dispensing in non-White communities: a 10% increase in the non-White proportion of the population yields a 3.39% (1.33%) increase in the overall opioid dispensing by pharmacies of high (average) supply chain complexity. Communities of color have been historically under-resourced and neglected by many government and social services (Potapchuk et al., 2005). In the

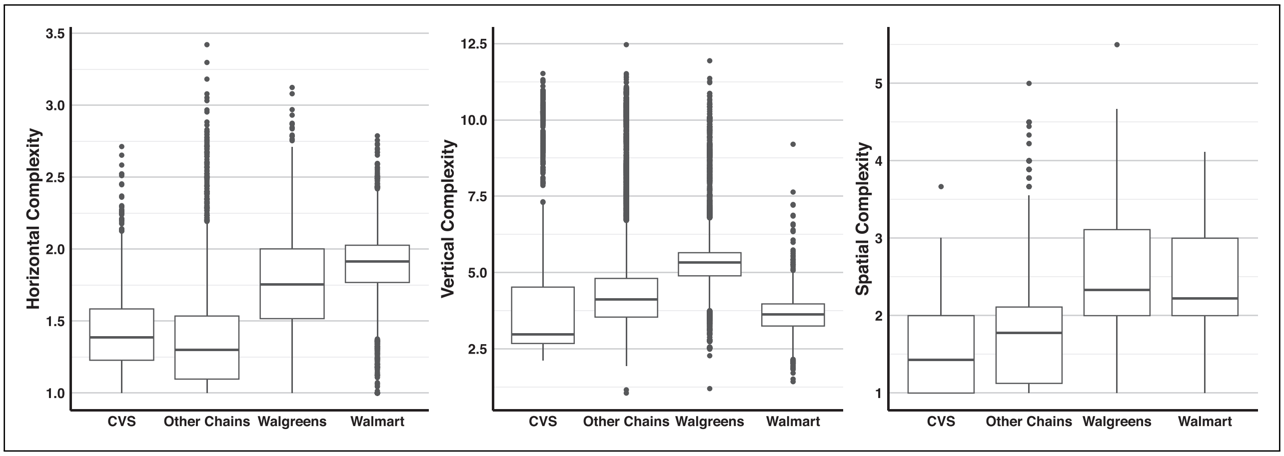

Figure 12. Distribution of supply chain complexity measures for major US pharmacy chains.

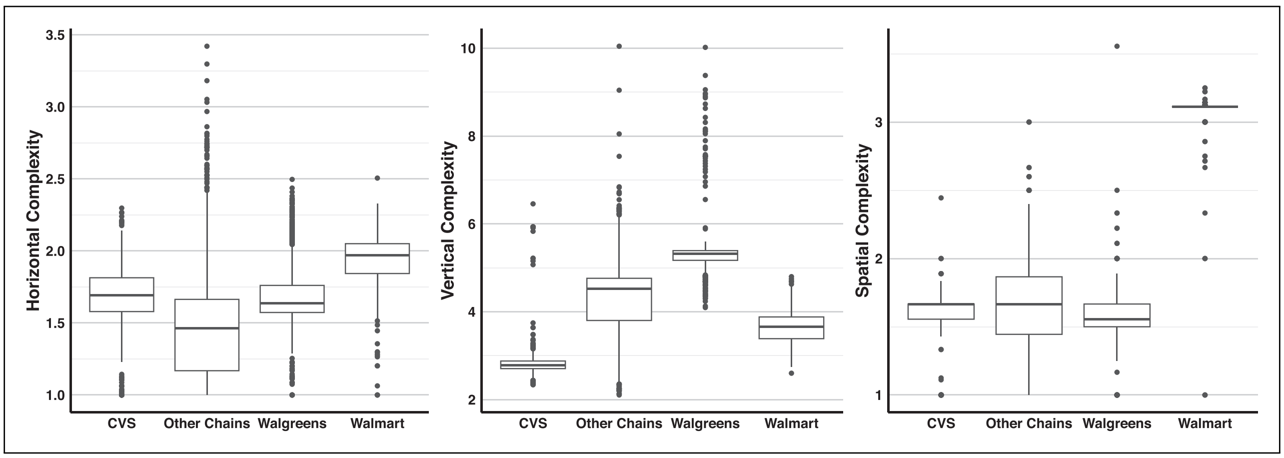

Figure 13. Distribution of supply chain complexity measures for major pharmacy chains located in Florida.

context of the opioid crisis, it appears that the DEA has spent more effort arresting non-White drug users (Fellner, 2009) than on regulating the flow of opioids from pharmaceutical companies into non-White communities. A recent report by the Office of the Inspector General suggests that insufficient staffing may be one of the problems: "[The] DEA is making an effort to increase both Diversion Investigator and Special Agent staffing levels in the field divisions hardest hit by the opioid epidemic," which are incorrectly believed to be mostly White communities (Office of the Inspector General, 2019). Our findings should increase awareness of racial bias in combating the opioid crisis and highlight that additional resources are needed to protect—rather than punish—communities that have long been considered unaffected by the opioid crisis.

6 Limitations and Future Work

Our study has several limitations. First, despite employing FEs, county-level controls, and exploiting an exogenous shock in a DiD model, our results should not be interpreted as causal due to the nature of the secondary data used. Although our analysis and supporting evidence from the DEA's own investigations suggest that the positive association between complexity and over-dispensing is likely attributable to pharmacies' attempts to avoid DEA monitoring and detection, it remains possible that some pharmacies working with smaller suppliers expanded their distribution networks to meet surges in demand. Furthermore, we find that a county's non-White proportion positively moderates the relationship between supply chain complexity and opioid dispensing. Although we argue that this is likely due to fewer monitoring resources being allocated to communities of color, an alternative explanation may relate to disparities in access to quality healthcare for non-White populations. Although we control for this by including socioeconomic factors and the number of treatment facilities in our estimations, exploring the potential impact of stigma and mistrust in the healthcare system—and its resulting contribution to more severe opioid outcomes—among communities of color offers a compelling avenue for future research.

Second, our analysis assumes that pharmacies within a county mainly serve local patients. Although this may not always be the case, this concern should be minimal because we focus on retail pharmacies, excluding pain clinics and "pill mills" that are more likely to serve patients from disparate geographies. Our data does not permit exploring different patient groups, which may offer a more nuanced understanding of disparities, and we believe this is an exciting opportunity for future work. Finally, providing direct evidence of DEA oversight deficiencies in non-White communities is challenging due to limited public data on staffing and resource allocation across divisions. This area is also promising for future work. Regardless of its limitations, we hope our work will spur further investigation of this topic within the operations management community.

Declaration of Conflicting Interests

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Supplemental Material

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Notes

1. We identified substance abuse treatment facilities by following the approach used by Liu and Bharadwaj (2020). Specifically, we included outpatient mental health and substance abuse centers (NAICS code: 621420), psychiatric and substance abuse hospitals (NAICS code: 622210), and residential mental health and substance abuse facilities (NAICS code: 623220).
2. $100 \times [exp(0.021) - 1] = 2.12\%$.

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